ExtremaFuse for Adaptive Compression of Frame-Level Speech Representations

Anonymous ACL submission

Abstract

How much can a recognizer shorten a speech sequence while retaining the distinctions needed to recognize phones? ExtremaFuse selects fusion centers at peaks and valleys of the cosine distance between adjacent frames and learns how to weight the five frames around each center. We apply it to Mel features and again inside an offline Conformer transducer trained on 460 hours of LibriSpeech. On test-clean, this system has a phone error rate of 5.69 percent, compared with 5.74 percent for a stride-four baseline with the same input projection width and a wider encoder. Relative to fusion at the Mel input alone, the second stage removes 59.8 percent of the remaining steps at a cost of 0.11 percentage points in phone error. We compare computation using major multiply–accumulate operations per second of input audio. Measured fusion weights show how neighboring frames still contribute to the shorter sequence.

1 Introduction

A stride-four encoder represents every four input frames with one time step, whether those frames cover a steady vowel or straddle a transition. A phone recognizer offers a direct test of this choice. A system might emit a shorter sequence yet introduce deletions where a brief sound becomes hard to distinguish. The arithmetic cost also depends on which encoder layers process the shortened sequence and how many candidate hypotheses the decoder scores.

We propose ExtremaFuse, a local operation on a sequence of frame vectors. Peaks and valleys in adjacent-frame distance determine where to emit a vector. The five frames around each selected position determine its value through learned weights. The operator can act on acoustic features or intermediate encoder states and takes no phone boundaries as input.

We ask two questions. First, within the computation budget of a fixed-stride recognizer, what phone error rate can the adaptive system achieve? We compare systems that match the per-frame projection or the encoder width and count major multiply–accumulate operations along the actual decoding path. Second, what does an additional round of compression cost after the Mel sequence has already been shortened? We count the steps and phone errors, inspect which input frames contribute to an output, and test a length-matched replacement in the same trained model. A breakdown by speaking rate reveals which edit types account for the difference in aggregate error.

2 Related Work

Time reduction for speech models. Funnel pooling reduces the frame rate inside large speech encoders, and fixed-stride subsampling has been studied at both the input and the output of Whisper (Wang et al., 2024; Cho et al., 2026). Adaptive methods also exploit neighboring representations. A-ToMe merges adjacent transducer tokens based on the similarity of their attention keys (Li et al., 2023). In audio language models, Bhati et al. (2025) compare segmentation from adjacent-frame distances with uniform pooling, and Xiang et al. (2026) study similarity-based merging at multiple depths. ExtremaFuse uses extrema in the distance curve as centers for overlapping, learned windows. This design lets us inspect which frames actually contribute to each output.

Local acoustic change and phone evaluation. Changes between neighboring features can provide cues for phonetic segmentation (Kreuk et al., 2020; Yantosca and Cheng, 2025). Here the selected positions serve as fusion centers. Their value is assessed through recognition of the reference phone sequence. Work on phone realization illustrates the range of distinctions hidden by a single aggregate score (Bharadwaj et al., 2026). We therefore also examine substitutions, deletions and insertions.

Input	$H = (h_1, \ldots, h_T)$
Select	$H \longmapsto (d_1, \ldots, d_{T-1}) \longmapsto C$
Fuse	$(H, C) \longmapsto (z_{c_1}, \ldots, z_{c_K})$

Figure 1: ExtremaFuse at one insertion site. Distances determine an ordered set C of centers. Each z_c pools the original features around that center with learned weights. The next stage receives K frames of dimension D.

3 ExtremaFuse

3.1 Sequence interface

The input is an ordered sequence $H = (h_1, \ldots, h_T)$ of valid D-dimensional frames. The output has the same feature dimension and a content-dependent length $K \leq T$. Downstream layers consume the new length and mask. Figure 1 shows how the original frames enter both selection and fusion.

3.2 Selecting fusion centers

For adjacent nonzero frames, define $d_t = 1 - \cos(h_t, h_{t+1})$. A peak at p selects frame $p + 1$, immediately after a locally large change in direction. A valley at v selects frame v at a locally small change. We merge duplicate positions and keep them in chronological order. If a nonempty sequence has no interior extremum, we use its first frame. The number of centers follows from the input, without a target retention rate.

3.3 Learning local fusion weights

For a center c, let W_c contain its valid neighbors up to two positions away, including c. A two-layer network takes the five concatenated frame vectors, applies a 32-unit ReLU layer and predicts five logits. Inputs outside the sequence use the nearest endpoint for this prediction. The softmax weights are renormalized over valid positions for the output,

$$z_c = \sum_{j \in W_c} \alpha_{cj} h_j, \quad \alpha_{cj} \geq 0, \quad \sum_{j \in W_c} \alpha_{cj} = 1. \tag{1}$$

Overlapping windows allow one input to contribute to more than one output. The center-selection rule has no trainable parameters. Each insertion site learns its own fusion weights through the recognition objective.

4 Experimental Setup

4.1 Task and systems

We train on LibriSpeech train-clean-100 and train-clean-360 (460 hours) and recognize sequences from a 69-phone ARPABET inventory with vowel stress (Panayotov et al., 2015). We evaluate ExtremaFuse in a ten-layer offline Conformer with a transducer objective (Gulati et al., 2020; Graves, 2012). In the two-stage system (E10), fusion first shortens the 80-dimensional Mel sequence before its input projection. A second module acts after layer 5, so the final five encoder layers receive the twice-compressed sequence.

Four-frame stacking couples the input projection and encoder width in our baselines. E09 matches E10's 128-dimensional per-frame projection and stacks four projected vectors, giving it a 512-dimensional encoder. E01 uses a 32-dimensional projection to match E10's 128-dimensional encoder and approximate its parameter count (Table 1). We train with seed 42 and select checkpoints by dev-clean validation loss. Batch budgets and update counts differ across runs. Appendix A records the updates at selection and at the end of training.

4.2 Uniform pooling control

Uniform pooling tests whether comparable compression at the same sites suffices. The planned control uses E10's backbone and replaces both fusion modules with means over contiguous, disjoint segments of as nearly equal length as possible. At each site, its output length depends only on its input length and a retention ratio calibrated from E10 on dev-clean. The rest of the recognizer will be trained from scratch with these two ratios fixed.

4.3 Quality, cost, and uncertainty

Corpus phone error rate (PER) divides total substitutions, deletions and insertions by the number of reference phones. We use the same 2,620 test-clean utterances for every system (189,023 phones, 40 speakers). Paired intervals resample speakers jointly across systems and recompute corpus PER on each of 10,000 draws. Test-clean informed the choice of configurations, so comparisons on this split are exploratory.

We measure computation by major multiply–accumulate operations (MACs) per second of input audio. For utterance counts C_i and actual audio durations d_i, the reported value is $\sum_i C_i / (10^9 \sum_i d_i)$. One MAC denotes one multiply–accumulate. We

System	Input proj.	Encoder width	Params (M)	test-clean PER (%)	test-clean GMAC/s	test-other PER (%)	test-other GMAC/s
Fixed stride 4 (E09)	128	512	29.58	5.7427	–	–	–
ExtremaFuse, Mel + layer 5 (E10)	128	128	4.25	5.6882	–	–	–
Fixed stride 4 (E01)	32	128	4.21	6.1289	0.24404	–	–
ExtremaFuse, layer 5 (E06)	128	128	4.24	4.7809	0.62512	–	–
Uniform pooling, Mel + layer 5	128	128	–	–	–	–	–

Table 1: Phone recognition quality and major MACs per second of input audio (GMAC/s), including encoder, compression, predictor and joiner operations. E09 matches E10's input projection; E01 matches its encoder width. Both metrics are lower-is-better. Dashes mark pending full-corpus measurements. E09-relative computation budgets remain pending until its count is complete.

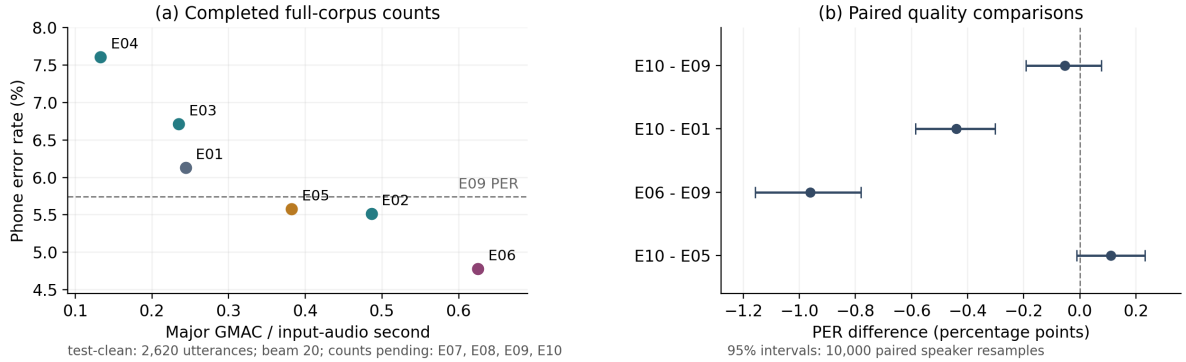


Figure 2: Test-clean PER and full-corpus computation counts for E01–E06 (left); E07–E10 counts are pending in this snapshot. The horizontal line marks E09's PER, with its computation coordinate still unreported. Paired PER differences in percentage points (right) use 10,000 paired speaker resamples.

use FP32, batch size one and beam width 20. Counts follow each layer's actual sequence length and the hypotheses expanded by beam search, with every output checked against the archived prediction.

The scope includes linear and convolution operations, both attention matrix products, fusion networks, adjacent-frame dot products, weighted fusion, predictor gates and joiner projections. We separate encoder and compression costs from predictor and joiner costs. Feature extraction, normalization, softmax, activations, extrema comparisons, sorting and memory access are outside this major-MAC scope. For the position study, a candidate meets the computation budget when its count divided by E09's count on the same split and decoding protocol is at most one. This ratio is pending until both full-corpus counts are available.

5 Results

5.1 Recognition and computation

The two-stage system (E10) has a test-clean PER of 5.6882%, compared with 5.7427% for E09. E10 makes 103 fewer errors, a difference of -0.054 percentage points with a paired 95% interval of $[-0.192, +0.076]$. The interval leaves the quality ordering uncertain. These complete systems also differ in encoder width, and their full-corpus computation comparison remains pending.

Matching the encoder width gives a second comparison. Relative to E01, E10's PER difference is -0.441 percentage points, with a paired 95% interval of $[-0.586, -0.303]$. The input projections and training budgets differ. E06 provides a higher-quality operating point at 4.7809% PER. Its full-corpus count is 0.62512 GMAC per audio second, compared with 0.24404 for E01 (Figure 2).

The completed counts also distinguish encoder savings from candidate-scoring work. Moving single-stage fusion from layer 5 (E06) to layer 2 (E02) reduces encoder and compression MACs from 0.43193 to 0.29744 GMAC per audio second, a 31.14% reduction. Predictor and joiner costs are 0.19319 and 0.18893, respectively, so the total falls by 22.20%, from 0.62512 to 0.48637. PER increases from 4.7809% to 5.5141%. E04 uses 0.13298 GMAC per audio second at 7.6081% PER. The remaining counts are needed before ranking

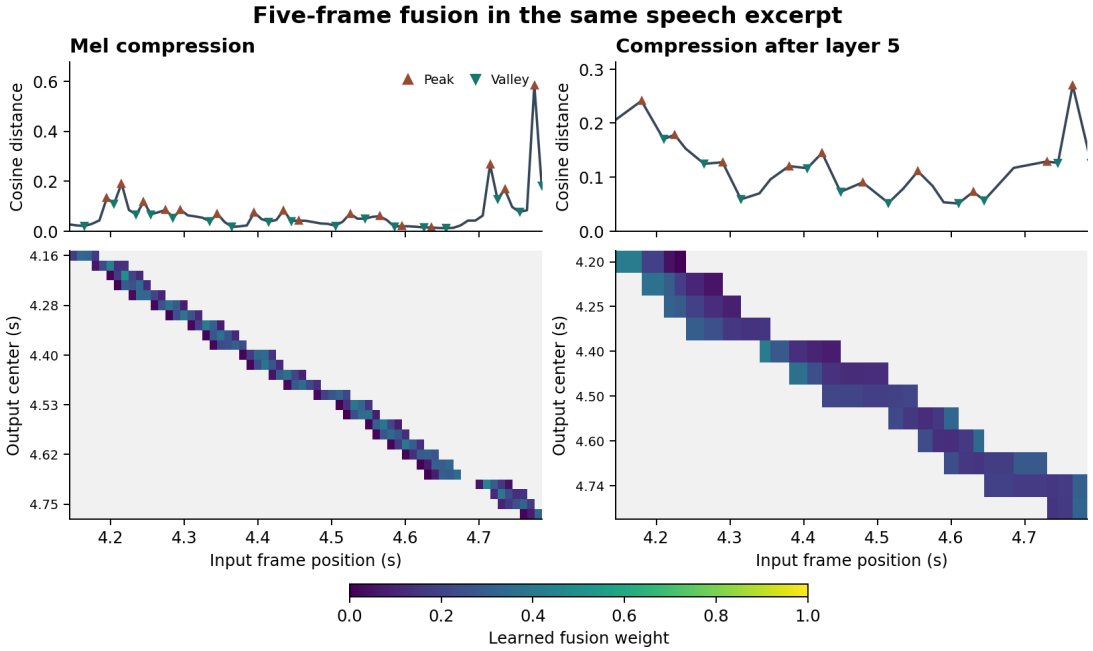


Figure 3: Measured distance curves and fusion weights for the same excerpt at the Mel input and after layer 5. Each row in the lower panels represents an output center. Layer-5 positions are expressed on the Mel time axis; their features already incorporate encoder context.

all systems under E09's computation budget.

5.2 Independent evaluation and pooling

Two comparisons remain open. On test-other we will test whether E10 stays within a prespecified acceptable PER increase over E09 and report the corresponding full-corpus MAC ratio. We will also compare E10 with independently trained uniform pooling at the same two sites and report realized frame counts, PER and major MACs. The empty entries in Table 1 are reserved for these measurements.

6 Analysis

6.1 The cost of the second fusion stage

To examine the second compression itself, the relevant reference is the Mel-only system (E05). Across test-clean, both E05 and E10 produce 764,950 Mel-stage outputs. E10 reduces this to 307,404 steps after layer 5, removing a further 59.81%. The final sequence retains 15.79% of the original Mel frames. E05's full major-MAC count is 0.38207 GMAC per audio second; E10's corresponding count is pending.

E10 makes 208 more errors than E05. PER increases from 5.5782% to 5.6882%, a difference of $+0.110$ percentage points (paired 95% interval $[-0.011, +0.232]$). These are separately trained systems. To test how much earlier the additional reduction can occur under the same computation budget, Appendix B.1 reserves one paired comparison with the second stage after layer 2.

6.2 Window coverage and a matched-length intervention

Equation 1 gives a bound on local distortion,

$$\|z_c - h_c\|_2 \le \sum_{j \in W_c} \alpha_{cj} \|h_j - h_c\|_2. \qquad (2)$$

When the neighbors receiving high weights resemble the center, the fused vector stays close to it. This gives a local explanation that can be checked against the actual inputs and weights. In 100 test-clean utterances (ten per speaking-rate decile, 37 speakers), adjacent vectors at layer 5 have a mean cosine similarity of 0.877. Pairs separated by half an utterance under a circular shift have a mean similarity of 0.143.

On this sample the second stage emits 40.19% as many vectors as it receives. Its windows cover 98.24% of the input positions, and 97.03% receive a weight of at least 0.05 in at least one output. The effective number of inputs per output, $1/\sum_j \alpha_{cj}^2$, averages 3.89. Figure 3 shows how a frame between two centers can contribute to both. Coverage measures participation in fusion. Phone errors quantify which distinctions survive recognition.

Group	Phones	ΔS	ΔD	ΔI	ΔPER
Slow	36,624	+55	−2	−40	+0.035
Middle	83,207	+95	−326	+279	+0.058
Fast	69,192	+85	−504	+255	−0.237

Table 2: E10 minus E09 on test-clean by speaking rate. S, D and I count substitutions, deletions and insertions. PER differences are in percentage points.

Could an ordinary mean produce a similar result once the encoder has supplied context? We freeze E10, keep its Mel module, and replace only the layer-5 module with means over equally spaced segments. For every utterance we use exactly the number of outputs that E10 selected. Reference-transcript negative log-likelihood changes from 0.1423 to 0.1435 nats per phone (difference $+0.00117$, paired 95% interval $[-0.00241, +0.00499]$). Thus the frozen model is also compatible with simple local averaging on this sample. The independently trained two-site control in Table 1 tests whether adaptive fusion improves recognition or decoding cost at a similar frame budget.

6.3 Speaking rate and the error profile

The small aggregate PER difference hides a change in error type. Across test-clean, E10 makes 832 fewer deletions than E09, alongside 235 more substitutions and 494 more insertions. To locate these changes, we group utterances by reference phones per minute of audio, including pauses. The slow and fast groups each contain three deciles and the middle group contains four.

The fast group accounts for 164 fewer errors, while the slow and middle groups contribute 13 and 48 additional errors (Table 2). E10 retains similar proportions of the original Mel frames across the three groups (15.74%, 15.77% and 15.84%). The fast-group PER difference is -0.237 percentage points (95% interval $[-0.447, -0.010]$). Testing whether the benefit is larger in fast speech requires comparing groups directly. The fast-minus-slow difference is -0.273 percentage points with an interval of $[-0.697, +0.154]$, leaving the rate dependence uncertain.

7 Conclusion

On LibriSpeech test-clean, applying ExtremaFuse at the Mel input and inside the encoder gives a phone error rate close to the stride-four baseline. The second stage cuts the remaining sequence by more than half. Its overlapping windows still assign weight to most input positions, and the frozen model tolerates replacing them with length-matched averages on the diagnostic sample. The completed counts show that moving fusion earlier reduces encoder work while candidate scoring limits the total reduction. Full counts for the primary two-stage comparison and an independently trained uniform-pooling control remain necessary to establish its computation benefit.

Limitations

Our evidence covers English read speech, phone recognition, one offline Conformer transducer and one training seed. The fixed-stride and adaptive systems differ in capacity or training exposure. Major MACs cover the disclosed arithmetic scope and depend on the input lengths and beam-search path. They omit scalar operations and memory traffic and do not establish wall-clock performance. E07–E10 full counts are pending in this draft. The speaker intervals quantify test-sample uncertainty for these checkpoints. Test-clean was used to develop the comparisons, and the reference transcripts supply no aligned phone boundaries with which to score the selected centers.

Ethical Considerations

The evaluation uses English audiobook recordings and stressed phone labels. Application to other speaking styles or accents would require corresponding evaluation data.

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A Configurations and Reproducibility

Table 3 gives the configurations behind Figure 2. For E09 the validation loss selected an earlier checkpoint, at 23,150 of 117,500 updates. E10 was resumed with an increased phone-token batch budget and selected at 89,901 of 90,986 updates. Test scores throughout the paper correspond to the selected checkpoints.

All systems use 16 kHz audio, 80 Mel bins and a 10 ms frame hop. Training uses Adam with a peak learning rate of 0.0005, 10,000 warm-up updates and gradient clipping at 10. The corpus bootstrap resamples 40 speakers with replacement using seed 42 and reports percentile intervals. All utterances from a sampled speaker enter each system's numerator and denominator together.

B Additional Diagnostics

B.1 Testing early placements under the computation budget

The stride-one, layer-2-only system (E02) uses 0.48637 GMAC per audio second on test-clean. Its E09-relative computation budget is pending. Table 4 reserves two early placements for distinct questions. A layer-1-only run would test a new one-stage candidate against both E02 and E09. A Mel-plus-layer-2 run could be compared with E02 to examine the addition of Mel fusion, and with E10 to examine moving the second module forward. In the latter comparison, layers 3 through 5 would also process the shortened sequence.

Both proposed runs use stride one and a 128-dimensional projection and encoder. Each would follow the corresponding reference training recipe, with the actual updates recorded. We would report stage input and output counts, paired PER and full-corpus major MACs, including any over-budget result. Candidate selection would use the E09-relative computation budget on dev-clean before confirmation on test-other.

B.2 Selection of the illustrated excerpt

For the weight diagnostic we select ten utterances per test-clean speaking-rate decile using a deterministic hash of the utterance identifier. Figure 3 shows a 0.64-second window centered at the midpoint of the lowest-hash eligible utterance in decile five (at least 200 raw frames). The choice therefore does not use recognition errors or the appearance of the weights.

ID	Compression site(s)	Stride	Proj./width	Selected step	Final step	PER (%)	GMAC/s
E01	Fixed stride	4	32/128	46,858	56,900	6.1289	0.24404
E02	Layer 2	1	128/128	69,137	126,550	5.5141	0.48637
E03	Layer 2	2	64/128	24,540	59,850	6.7135	0.23493
E04	Layer 2	4	32/128	43,524	48,550	7.6081	0.13298
E05	Mel	1	128/128	121,688	126,550	5.5782	0.38207
E06	Layer 5	1	128/128	151,700	165,000	4.7809	0.62512
E07	Layer 7	1	128/128	118,826	126,550	4.8603	–
E08	Keep at 2, fuse at 5	1	128/128	73,182	141,382	5.4210	–
E09	Fixed stride	4	128/512	23,150	117,500	5.7427	–
E10	Mel + layer 5	1	128/128	89,901	90,986	5.6882	–

Table 3: Full test-clean PER and completed full-corpus major-MAC counts per input-audio second; dashes mark pending counts. For E08 the first stage keeps the union of each selected center and its immediate neighbors. The second applies learned five-frame fusion. Stride is the fixed stacking factor.

Fusion site(s)	PER (%)	GMAC/s	Ratio/E09
Layer 2 (E02)	5.5141	0.48637	–
Mel + layer 5	5.6882	–	–
Layer 1 only	–	–	–
Mel + layer 2	–	–	–

Table 4: Early-placement comparisons on test-clean. PER is available for E02 and E10; the completed count is available for E02. Dashes mark pending measurements or optional experiments. A major-MAC ratio of 1 is the baseline computation budget.